GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Chemical Engineering

Subject Code: DI05005041

Subject Name: Nano Technology

w. e. f. Academic Year:	2026-2027
Semester:	V
Category of the Course:	PEC- III

Prerequisite:	Students should have basic knowledge of Physics (modern physics & electromagnetism), Chemistry (atomic structure & bonding), Mathematics (calculus), and fundamentals of Materials Science/Electronics to understand nanoscale concepts and fabrication techniques.
Rationale:	Nanotechnology is an emerging interdisciplinary field significantly impacting chemical industries, energy systems, healthcare, advanced materials and environmental engineering. Diploma Chemical Engineers are required to understand nanoscale phenomena, nanomaterial synthesis, characterization techniques, industrial applications and safety considerations. This course develops competency in nanotechnology fundamentals aligned with industrial requirements and cognitive learning outcomes.

Course Outcome:

After Completion of the Course, Student will able to:

No	Course Outcomes	RBT Level
01	Understand nanotechnology fundamentals and nanoscale concepts.	R, U
02	Explain types of nanomaterials, synthesis methods and properties.	R, U
03	Apply nanotechnology principles in energy, medicine and materials engineering.	R, U, A
04	Analyze industrial applications, benefits and limitations of nanotechnology.	U, A
05	Interpret safety, environmental and regulatory aspects of nanotechnology.	R, U

**Revised Bloom's Taxonomy (RBT)*

GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Chemical Engineering

Subject Code: DI05005041

Subject Name: Nano Technology

Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE(E)	PA(M)	PA(I)	ESE (V)	
3	0	2	4	70	30	20	30	150

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
Unit – I Introduction to Nanotechnology	1.1 Definition and historical development of nanotechnology 1.2 Concept of nanoscale (1–100 nm) and its significance 1.3 Unique properties at nanoscale 1.4 Scope and applications of nanotechnology	05	12
Unit – II Nanomaterials	2.1 Classification: 0D, 1D, 2D and 3D nanomaterials 2.2 Top-down and bottom-up synthesis approaches 2.3 Physical, chemical and optical properties 2.4 Surface area to volume ratio and quantum effects 2.5 Industrial relevance of nanomaterials	10	22
Unit – III Nanoparticles and Characterization	3.1 Types: metal, semiconductor and ceramic nanoparticles 3.2 Synthesis techniques: sol-gel, chemical reduction, precipitation 3.3 Characterization techniques: SEM, TEM, XRD, UV-Visible (basic concepts) 3.4 Applications in catalysis, electronics, sensors and coatings	10	22
Unit – IV Nanotechnology in Energy and Medicine	4.1 Nanotechnology in solar cells, fuel cells and batteries 4.2 Drug delivery systems and biomedical imaging 4.3 Nanotechnology in materials science: nanocomposites and coatings 4.4 Industrial case studies	08	18
Unit – V Safety and	5.1 Health and safety considerations in handling nanomaterials	06	14

Environmental Aspects	5.2 Toxicity and environmental impact of nanoparticles 5.3 Regulatory guidelines and safe disposal practices		
Unit – VI Recent Trends and Emerging Applications	6.1 Recent advancements in nanotechnology 6.2 Emerging smart nanomaterials 6.3 Future scope and industrial trends	06	12
	Total	**45**	**100**

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
20%	**45%**	**35%**	---	———	———

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

References/Suggested Learning Resources:

(a) Books:

S. No.	Title of Book	Author	Publication with place, year and ISBN
1	Introduction to Nanoscience and Nanotechnoly	K.K. Chattopadhyay & A.N. Banerjee	PHI Learning.
2	Introduction to Nanotechnology	.P. Poole Jr. & F.J. Owens	Wiley.
3	Nano: The Essentials	T. Pradeep	McGraw-Hill Education.

(b) Open-source software and website:

Relevant NPTEL / SWAYAM online courses.

Suggested Practical List:

Sr. No.	Practical/Exercise (Course Outcomes in Psychomotor Domain according to NBA Terminology)	Approx. Hrs Required
1	Aim: To understand nanoscale dimensions (1–100 nm).	2
2	Aim: To demonstrate increase in surface area with decrease in particle size.	2

3	Aim: To prepare nano-sized powder using ball milling or mortar-pestle grinding (demonstration).	2
4	Aim: Preparation of silica nanoparticles using sol-gel process.	2
5	Aim: Observe color change in colloidal nanoparticles (e.g., gold/silver nanoparticles).	2
6	Aim: To synthesize Ag nanoparticles using chemical reduction.	2
7	Aim: To prepare ZnO nanoparticles using precipitation technique.	2
8	Aim: To study surface morphology using Scanning Electron Microscope.	2
9	Aim: To understand internal structure analysis using Transmission Electron Microscope.	2
10	Aim: To study crystal structure using X-ray diffractometer.	2
11	Aim: To determine absorption peak using UV-Visible spectrophotometer.	2
12	Aim: To understand nanotechnology in solar cells.	2
13	Aim: Preparation of polymer nanocomposite.	2
14	Aim: To understand targeted drug delivery concepts.	2
15	Aim: To understand environmental impact testing procedures.	2
16	Aim: To understand emerging nanomaterials.	2

Suggested Activities for Students:

1. Assignments
2. Technical Quiz/MCQ Test
3. Synthesis & testing of nano-coated antibacterial fabric
4. Nano-adsorbent for water purification
5. Conductive nanocomposite preparation
6. Dye-sensitized solar cell prototype
7. Nano-sensor concept design
8. Preparation of seminar on recent trends
9. Industry interaction / expert lecture

Case Study Discussions

- Nano drug delivery systems
- Nano solar cells
- Carbon nanotubes in electronics
- Environmental nano-filters

* * * * * * *